

Tie Strength, Role Relations, and Social Support in Ego Networks

David Hachen* and Omar Lizardo†

September 2026

Abstract

Network analysts often treat tie strength as a single continuum or conflate it with role relations and resource provision. Drawing on multi-wave panel data from the NetHealth Study ($N = 22,739$ ego–alter tie observations across $N = 626$ undergraduate participants), we untangle the multidimensional structure of tie strength, social roles, and social support. First, using Latent Class Analysis (LCA) across four support exchanges (companionship, advice, comfort, and financial aid), we identify four distinct relational configurations: “Casual Companionship,” “Comprehensive Support,” “Instrumental Support,” and “Peripheral Support.” Second, using multilevel generalized linear mixed models with random ego intercepts, we predict membership in each inductively identified latent support configuration based on tie-strength indicators, role relations, and contextual attributes, while accounting for dyadic clustering. Results show that emotional closeness, relationship duration, and cognitive salience strongly sort ties into comprehensive support, whereas peer friendships dominate casual companionship. Furthermore, role relations condition support regimes independently of tie strength: family ties specialize in instrumental support, whereas romantic partners provide high levels of comprehensive support. Men’s networks are disproportionately weighted toward casual companionship rather than comprehensive support. These findings show that personal communities are organized through a functional division of relational labor rather than an undifferentiated gradient of tie strength.

Keywords: tie strength; social support; ego networks; latent class analysis; multilevel GLMM; role relations; NetHealth study.

*Department of Sociology, University of Notre Dame, Los Angeles. Email: dhachen@nd.edu.

†Department of Sociology, University of California, Los Angeles. Email: olizardo@soc.ucla.edu.

1 Introduction

The concept of tie strength occupies an indispensable role in structural sociology, social network analysis, and theories of social exchange (Borgatti et al., 2009; Granovetter, 1973; Kitts, 2014; Marsden and Campbell, 1984). In his seminal formulation, Granovetter (1973, p. 1361) defined tie strength as a “combination of the amount of time, the emotional intensity, the intimacy (mutual confiding), and the reciprocal services which characterize the tie.” While this multidimensional definition highlighted the multifaceted nature of interpersonal bonds, it also blended distinct conceptual domains into a single construct, conflating subjective sentiments (emotional closeness), observable behaviors (interaction frequency), relational history (duration), and functional exchanges (provision of tangible assistance and social support).

In an influential methodological intervention, Marsden and Campbell (1984) clarified this conceptual ambiguity by distinguishing between *indicators* and *predictors* of tie strength. Indicators represent the internal properties that constitute the strength of the bond itself, such as subjective emotional closeness, frequency of communication, and relationship duration. Predictors, by contrast, represent the structural and social contexts that generate or sustain strong ties, including social role relations (such as kin, friends, romantic partners, and acquaintances), sociodemographic similarity, and shared institutional affiliations (Feld, 1982). Despite this conceptual clarification, empirical scholarship has frequently slipped back into definitional conflation. In empirical applications, researchers often assume that kin and close friends are “strong ties” by definition, while neighbors, coworkers, and acquaintances are assumed to be “weak ties” (Lin, 2001; Wellman and Wortley, 1990). This conflation obscures crucial empirical variation: individuals routinely maintain emotionally distant or infrequently contacted relatives alongside intensely close and supportive casual associates.

As Lizardo (2024) argues in a recent theoretical formulation, this persistent conflation stems from a deeper failure to recognize that the concept of a social tie is organized across multiple distinct cognitive and relational frames that define the different attributes definitional of the concept for researchers. Specifically, tie conceptualizations routinely collapse four autonomous tie attributes: *role type* denoting categorical social positions (such as kin, friend, or coworker), *sentiments* capturing affective evaluations (such as closeness, liking, or trust), *behavioral interactions* indexing contact and communication patterns (such as daily or weekly activation), and *exchange and support* encompassing the functional transmission of resources (such as companionship, advice, comfort, or financial assistance). Because frames are recursive of these attributes (like closeness, interaction frequency, or the exchange of favors) for instance, they are bundled into a “meta” attribute that is usually referred to as “strength,” as per

Granovetter (1973)’s classic definition. However, when researchers treat one attribute as an automatic index of another—such as assuming that role relations define tie strength, that high interaction frequency entails emotional intimacy, or that strong ties universally provide all forms of social support—they commit a concept-confounding mistake that can obscure the underlying mechanisms operating at the relational level in personal networks. Role relations may carry institutional expectations of solidarity, yet actual sentiments and behavioral activations vary substantially across role categories (Marsden and Campbell, 1984). Similarly, frequent interaction is often driven by situational co-presence in shared organizational foci rather than affective depth (Feld, 1982; Kitts, 2014), while the provision of specialized support is bounded by domain-specific norms and resource affordances rather than by a generalized gradient of strength.

To make headway on these issues, this study presents an empirical investigation that systematically disentangles tie-strength indicators, social role relations, and multidimensional support exchanges across personal networks. Analyzing multi-wave ego-network panel data from the NetHealth Study, which tracked undergraduate students across collegiate semesters at the University of Notre Dame, we use two analytical methods tailored to this structural complexity. Rather than imposing predetermined heuristic categories or assuming an additive hierarchy of support, we first use Latent Class Analysis (LCA) across four primary support indicators (companionship, advice, comfort, and financial assistance) to inductively identify empirical configurations of support. We then specify multilevel generalized linear mixed models (GLMMs) with a random ego intercept that directly predict membership in each identified latent support configuration from emotional closeness, interaction frequency, cognitive salience, and duration, while conditioning on social role relations, ego gender identity, alter attributes, and residential proximity. By bridging inductive latent typologies with confirmatory multilevel regressions, this framework provides a rigorous structural foundation for evaluating how personal communities coordinate intimacy, interaction, and mutual assistance without definitional circularity.

2 Theoretical Background and Research Questions

Social ties serve as the fundamental conduits through which individuals access psychosocial resources, instrumental help, and novel information (Borgatti et al., 2009; Lin, 2001). Building on the structural taxonomies proposed by Borgatti et al. (2009) and Kitts (2014), interpersonal connections can be characterized across four analytical planes: role relations (formal social positions such as family, friend, romantic partner, or coworker), interpersonal sentiments (affective closeness, trust, and liking), behavioral interactions (communication frequency and

shared activities), and resource exchanges (the provision of informational, emotional, and material support).

Marsden and Campbell (1984) measurement model established that emotional closeness serves as the primary and most valid empirical indicator of tie strength, whereas contact frequency functions as an imperfect indicator that is heavily contaminated by institutional and organizational constraints. For example, coworkers or dormmates may interact daily due to physical proximity rather than mutual confiding, whereas geographically separated kin may interact infrequently while maintaining profound emotional intimacy. Similarly, cognitive salience—the cognitive accessibility of an alter as reflected in the order in which they are retrieved during name generation—reflects psychological prominence that may diverge from sheer contact volume (Kitts, 2014). Finally, relationship duration reflects the temporal accumulation of relational investments, yet whether longevity translates into active support depends heavily on role context.

As Smith (2021) notes in a recent systematic synthesis of ego-network theory and research, the measurement of social support represents the single most common substantive application of personal network data (Cornwell et al., 2008; Fischer, 1982; Wellman and Wortley, 1990). Yet empirical investigations in this tradition often rely on aggregate summary measures—such as total network size or simple degree counts of available supporters—treating personal networks as an undifferentiated reservoir of assistance. As Smith (2021) argues, this aggregate approach obscures the reality that different alters offer fundamentally different resources: some provide material aid (financial assistance), others offer informational guidance (advice), and others supply expressive support (emotional comfort) or everyday sociability. Because ego network surveys collect detailed information on the specific nature of each dyadic bond—including emotional closeness, contact frequency, duration, and role category—they provide the empirical resolution required to examine how distinct dimensions of tie strength activate specific functional exchanges. Decoupling these dimensions at the dyadic level reveals how personal communities organize a specialized division of relational labor, rather than assuming that strong ties universally provide all forms of aid.

These conceptual distinctions motivate several core empirical questions regarding how tie strength and role relations interact to structure social support provision:

First, how do the empirical indicators of tie strength correlate across different social role relations? If emotional closeness and interaction frequency represent partially autonomous dimensions, we should observe distinct tie-strength profiles across roles. In particular, family ties are expected to exhibit high emotional closeness despite lower everyday activation frequency compared to peer relationships on a residential campus.

Second, what empirical configurations of social support characterize personal networks

when evaluated without arbitrary heuristic binning? Traditional analyses often assume an additive hierarchy of support, presuming that strong ties provide all forms of support while weak ties provide none. By applying Latent Class Analysis, we examine whether support exchanges cluster into specialized functional bundles, such as purely expressive support, casual socializing, or comprehensive multi-domain aid.

Third, when accounting for the clustering of ties within egos, how do emotional closeness, interaction frequency, and duration predict membership in distinct support configurations? We expect emotional closeness to serve as the primary determinant of comprehensive multi-domain support, whereas interaction frequency should primarily sustain casual companionship.

Fourth, to what extent do role relations condition support configurations independently of tie strength? Certain resources, particularly financial assistance, involve material costs and institutional norms of obligation that may be restricted to specific role categories (such as family or romantic partners), regardless of how frequently an alter is contacted or how emotionally close they are reported to be.

3 Data and Analytical Sample

The empirical data for this study come from the NetHealth Study (e.g., [Liu et al., 2018](#); [Sepulvado et al., 2020](#); [Wang et al., 2020](#)), a longitudinal investigation tracking an entire undergraduate cohort at the University of Notre Dame from matriculation in August 2015 through graduation in May 2019. The NetHealth study was designed to investigate the reciprocal co-evolution of social network structures, health behaviors (physical activity, sleep patterns, stress), and academic performance. Network surveys were administered online via Qualtrics at the beginning and conclusion of each semester across repeated waves between Fall 2015 and Spring 2018.

In each survey wave, participants completed an egocentric network module using a standardized open-ended name generator:

“We would like to know who you consider to be in your social network. In the spaces below, please list up to 20 people with whom you spend time communicating or interacting.”

Beginning in Wave 3, respondents could additionally retain up to five alters from the preceding wave, allowing for up to 25 named alters per wave. Following name generation, respondents completed detailed name interpreters for each listed alter, recording demographic attributes, relationship type, emotional closeness, contact frequency, relationship duration, and residential proximity. Crucially, across Waves 2, 3, 4, 5, 7, and 8, respondents were

administered a dedicated social support module inquiring whether each nominated alter had provided four specific forms of assistance over the preceding months. The battery was introduced with the standardized instruction:

“Please check the box if the contact is somebody who has provided that type of support.”

Respondents evaluated each alter across four specific functional support items:

1. **Companionship:** *“Somebody you have hung out with when feeling like company”*
2. **Advice:** *“Somebody you went to for advice when dealing with a life problem”*
3. **Comfort:** *“Somebody you counted on to comfort you when something bad happened”*
4. **Financial Assistance:** *“Somebody you have gone to when in financial need”*

For egos who were administered the support battery, unendorsed items represent the documented absence of that support type. We restricted the analytical sample to egos who completed the support module in each respective wave and eliminated records with incomplete covariate data. The final analytical sample consists of $N = 22,739$ ego–alter tie observations nested within $N = 626$ unique respondents across the six support-administered waves.

Tie strength indicators were coded following standardized survey categories: emotional closeness was recorded on a four-level ordinal scale (“Distant,” “Less than Close,” “Close,” and “Especially Close”); reported contact frequency was measured as “Less than Monthly,” “Monthly,” “Weekly,” or “Daily”; relationship duration was categorized into four temporal intervals (“Less than 1 Year,” “2–4 Years,” “5–10 Years,” and “More than 10 Years”); and cognitive salience was indexed both by exact nomination order (integers 1 through 20) and grouped into ordinal tiers (“Top 5,” “Middle 10,” and “Bottom 5”). Social role relations were standardized into five substantive categories: “Friend” (72.8% of nominations), “Family” (21.4%), “Romantic Partner” (2.8%), “Acquaintance” (1.4%), and “Other Role” (1.6%). Contextual controls include ego gender identity (women vs. men), alter gender, shared dormitory residence, and roommate status.

4 Latent Typologies of Social Support

To identify the underlying structure of social support provision without imposing arbitrary Boolean grouping heuristics, we estimated Latent Class Analysis models across the four binary support indicators in the R statistical computing environment (R Core Team, 2026)

using the `poLCA` package (Linzer and Lewis, 2011). We evaluated latent class solutions ranging from $K = 2$ to $K = 5$ classes. Table 1 reports the model fit statistics across the candidate specifications, including the log-likelihood, number of estimated parameters, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and classification entropy.

Table 1: Model Fit Statistics for Latent Class Analysis of Social Support Exchanges

Latent Classes	Log-Likelihood	Par	AIC	BIC	Entropy
2 Classes	-42,348.5	9	84,714.9	84,787.2	0.760
3 Classes	-41,670.9	14	83,369.8	83,482.2	0.858
4 Classes	-41,626.6	19	83,291.2	83,443.8	0.766
5 Classes	-41,626.6	24	83,301.2	83,493.9	0.722

Note: Model estimation based on $N = 22,739$ ego–alter tie observations across Waves 2, 3, 4, 5, 7, and 8 of the NetHealth Study. Par indicates the number of estimated parameters. Optimal fit is identified by the four-class solution (bolded) based on AIC and BIC minimization.

As reported in Table 1, moving from a two-class model to a three-class model yields a substantial reduction in both AIC ($\Delta\text{AIC} = -1,345.2$) and BIC ($\Delta\text{BIC} = -1,305.0$). Progressing from three to four classes provides a further improvement in model fit (AIC = 83,291.2; BIC = 83,443.8), while preserving high classification entropy (0.766). When moving to a five-class model, the BIC increases from 83,443.8 to 83,493.9 and the log-likelihood fails to improve, indicating parameter redundancy and overfitting. Consequently, the four-class solution was selected as the optimal empirical representation of social support configurations in this collegiate population.

From a methodological standpoint, the unit of analysis in this latent class model is the dyadic ego–alter tie observation ($N = 22,739$). Because ties are nested within respondents ($N = 626$), reports of social support exhibit non-trivial clustering at the ego level. With only four binary support indicators, however, there are $2^4 = 16$ observable response profiles (15 degrees of freedom in the saturated multinomial table). Estimating a simultaneous multilevel mixture model with continuous random effects across this compact discrete state space introduces severe empirical identification and convergence constraints. In our analytical design, we therefore implement a sequential, two-stage modeling strategy. In the first stage, we use single-level Latent Class Analysis strictly as an inductive, taxonomical tool to identify population-averaged relational configurations across dyads—replacing arbitrary manual groupings with data-driven empirical clustering. In the second stage, we classify each tie into its modal latent support regime and fit multilevel generalized linear mixed models with ego-specific random intercepts. This two-stage approach reserves all confirmatory hypothesis testing, predictor evaluation, and standard error estimation for the multilevel

regression framework, in which clustering within egos is explicitly modeled directly on the latent support configurations.

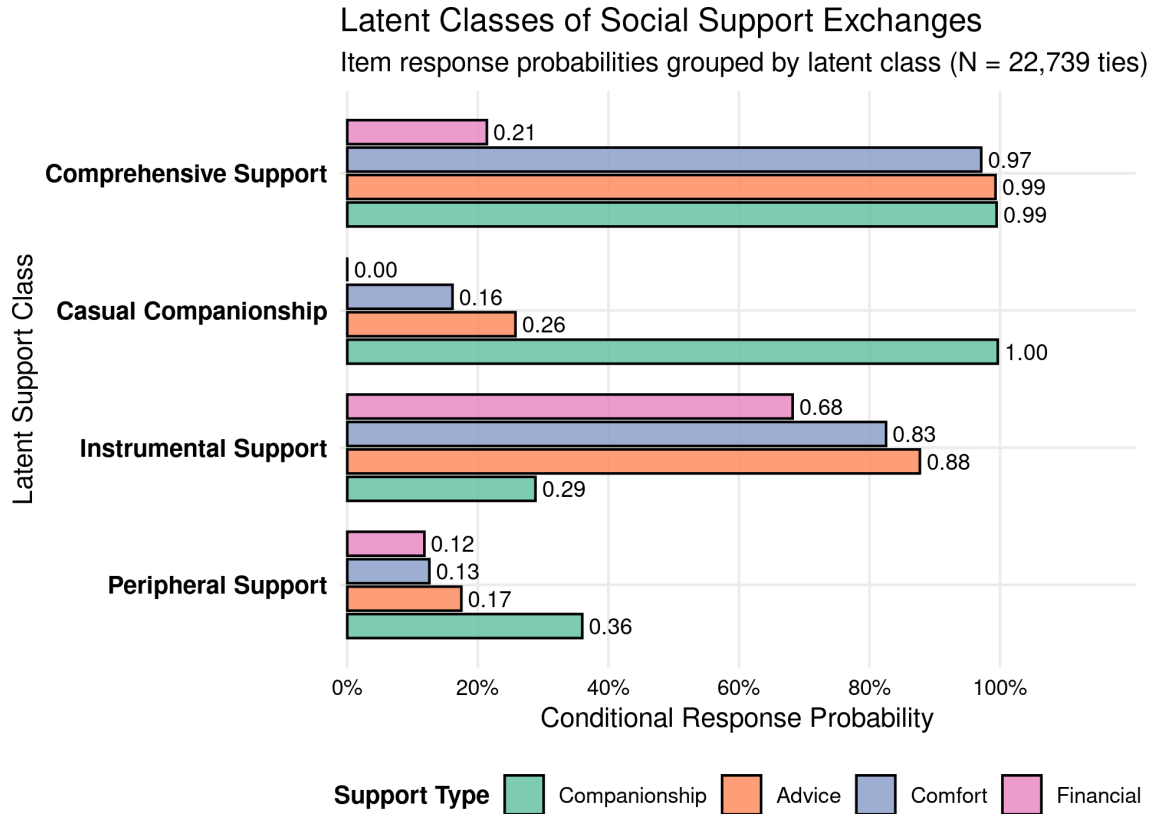


Figure 1: Conditional Item Response Probabilities Across Latent Social Support Classes

Note: Latent class profiles estimated via **poLCA** across four binary social support indicators ($N = 22,739$ tie observations). Bars represent the conditional probability of endorsing each support exchange given membership in that latent class, grouped by latent class and colored by support exchange item.

Figure 1 illustrates the conditional item response probabilities that define the four latent support classes, grouped by latent class and colored by support type. The first class, which we label “Casual Companionship,” accounts for 45.8% of all analyzed ties. Ties in this class have a 1.00 probability of providing companionship (hanging out) but a zero probability of providing financial assistance, with modest probabilities of providing advice (0.26) or comfort (0.16). This class reflects everyday collegiate socializing that lacks deep confiding or material exchange. The second class, designated “Comprehensive Support,” encompasses 41.5% of ties and represents emotionally intimate relationships characterized by high probabilities of advice (0.99), comfort (0.97), and companionship (0.99), as well as a moderate likelihood of financial aid (0.21). The third class, labeled “Instrumental Support” (4.3% of ties), displays a distinctive profile: it exhibits high probabilities of financial assistance (0.68), advice (0.88),

and emotional comfort (0.83), but a low probability of companionship (0.29). Finally, the fourth class, labeled “Peripheral Support” (8.4% of ties), exhibits depressed endorsement rates across all four domains, with companionship reaching only 0.36 and financial aid just 0.12.

Table 2: Distribution of Latent Support Classes by Social Role Relation and Emotional Closeness

Dimension & Category	Casual Companionship	Comprehensive Support	Instrumental Support	Peripheral Support
Panel A: Social Role Relation				
Friend	53.8%	40.5%	1.0%	4.8%
Family	15.3%	58.3%	17.8%	8.6%
Romantic Partner	14.8%	81.7%	1.5%	1.9%
Acquaintance	44.8%	5.4%	0.5%	49.3%
Other Role	50.5%	13.8%	2.2%	33.5%
Panel B: Emotional Closeness				
Distant	43.5%	16.5%	3.5%	36.5%
Less Close	66.5%	5.0%	1.6%	26.9%
Close	69.3%	19.3%	2.4%	9.0%
Especially Close	29.0%	61.7%	6.2%	3.1%

Note: Cell entries represent row percentages displaying the proportion of ties within each role or closeness category assigned to the respective latent support class ($N = 22,739$).

Table 2 presents the cross-classification of assigned latent support classes across social role relations (Panel A) and emotional closeness levels (Panel B). The empirical sorting reveals pronounced institutional and affective specialization. Among friends, ties are split predominantly between Casual Companionship (53.8%) and Comprehensive Support (40.5%), with virtually zero presence in Instrumental Support (1.0%). By contrast, family ties are heavily concentrated in Comprehensive Support (58.3%) and Instrumental Support (17.8%), while Casual Companionship accounts for only 15.3%. Romantic partners exhibit the highest concentration of Comprehensive Support (81.7%), with minimal peripheral categorization (1.9%). Acquaintances and other roles are primarily relegated to Peripheral Support (49.3% and 33.5%, respectively) and Casual Companionship.

Examining Panel B of Table 2 shows that emotional closeness strongly stratifies support configurations. Ties classified as “Especially Close” are overwhelmingly located in Comprehensive Support (61.7%), whereas ties categorized as “Close” or “Less Close” are predominantly confined to Casual Companionship (69.3% and 66.5%, respectively). Peripheral support is heavily concentrated among distant, less-close ties, indicating that emotional intimacy is an indispensable prerequisite for multi-domain support provision.

5 Multilevel Models: Predicting Latent Support Configurations

To test how tie strength indicators and role relations independently predict membership in each of the inductively identified support configurations while accounting for the nesting of multiple ties within respondents, we fit multilevel generalized linear mixed models (GLMMs) with a logit link function using the `lme4` package (Bates et al., 2015) in R (R Core Team, 2026). The outcome variables in this confirmatory stage are the four distinct relational regimes identified by the LCA: Comprehensive Support, Casual Companionship, Instrumental Support, and Peripheral Support.

For each latent support category k (Comprehensive Support, Casual Companionship, Instrumental Support, or Peripheral Support), the probability that a dyadic tie i nested within ego j is assigned to class k is modeled as:

$$\text{logit}(P(Y_{ij} = k)) = \beta_{0k} + \mathbf{X}_{ij}\boldsymbol{\beta}_k + u_{jk} \quad (1)$$

where $\mathbf{X}_{ij}$ represents the vector of tie-level strength indicators (affective closeness, contact frequency, cognitive salience, and duration), social role relations, and dyadic controls; $\boldsymbol{\beta}_k$ is the class-specific vector of fixed-effect coefficients, and $u_{jk} \sim \mathcal{N}(0, \sigma_{uk}^2)$ represents the ego-specific random intercept for class k . By capturing unobserved respondent-level variance—such as individual baseline sociability, reporting thresholds, or dispositional tendencies to perceive support—the inclusion of u_{jk} prevents standard error deflation and accounts for dyadic clustering within personal networks. Latent intraclass correlation coefficients (ICCs) demonstrate substantial ego-level clustering across all four configurations, ranging from 0.248 for Casual Companionship ($\sigma_u^2 = 1.082$) to 0.307 for Comprehensive Support ($\sigma_u^2 = 1.457$), 0.356 for Instrumental Support ($\sigma_u^2 = 1.821$), and 0.380 for Peripheral Support ($\sigma_u^2 = 2.016$).

From an estimation standpoint, specifying separate binary GLMMs for each latent configuration offers distinct methodological advantages over simultaneous multinomial mixed models (such as `mblogit`; Elff, 2025). In dyadic network data, certain cross-classifications between role relations and specialized support regimes contain extreme cell sparsity—most notably, acquaintances providing instrumental kin support ($n = 1$). Sparse cells induce quasi-complete separation, leading to non-convergence warnings and unstable covariance matrices in simultaneous multinomial estimators. The binary GLMM framework avoids these computational issues, preserves full categorical granularity across all role categories and duration intervals, and allows the respondent-level variance (σ_{uk}^2) to vary freely across distinct support regimes. Figure 2 visualizes the estimated adjusted odds ratios (ORs) and 95%

confidence intervals from these multilevel models across the four latent support configurations using `ggplot2` (Wickham, 2016).¹

Figure 2 visualizes the estimated odds ratios across the four latent social support configurations, allowing for direct comparison of how tie strength and role relations sort personal networks into distinct relational regimes. The parameter estimates demonstrate clear empirical patterns:

First, emotional closeness and long-term history serve as the primary engines sorting ties into Comprehensive Support. Relative to ties evaluated as “Close,” alters characterized as “Especially Close” exhibit more than a sevenfold increase in the odds of providing comprehensive support ($OR = 7.09, 95\% CI [6.46, 7.79]$), whereas “Less Close” ties show an 77% reduction in odds ($OR = 0.23, p < 0.001$). Relationship duration demonstrates powerful cumulative returns: ties maintained for 5–10 years ($OR = 2.10, p < 0.001$) and more than 10 years ($OR = 2.37, p < 0.001$) are more than twice as likely to provide comprehensive support compared to newer ties (2–4 years). Cognitive salience also enhances access (Top 5 : $OR = 1.36, p < 0.001$), while daily activation yields a modest elevation ($OR = 1.22, p < 0.001$). Net of tie strength, romantic partners are more than five times as likely as friends to occupy the comprehensive support regime ($OR = 5.20, 95\% CI [4.08, 6.63]$).

Second, Casual Companionship is overwhelmingly structured by peer friendships and moderate affective intimacy. Relative to friends, family members ($OR = 0.40, p < 0.001$), romantic partners ($OR = 0.21, p < 0.001$), and acquaintances ($OR = 0.21, p < 0.001$) are all substantially less likely to provide exclusively casual companionship. Interestingly, ties evaluated as “Especially Close” show sharply lower odds of remaining in this class ($OR = 0.24, p < 0.001$), because extreme affective closeness shifts ties into Comprehensive Support. Similarly, alters nominated in the Top 5 salience tier are less likely to be casual companions ($OR = 0.69, p < 0.001$), confirming that cognitive retrieval prioritizes more supportive confidants over recreational peers.

Third, Instrumental Support is governed by kinship roles rather than by campus proximity or daily interaction. Family ties exhibit a dramatic thirteen-fold elevation in the odds of providing instrumental support relative to friends ($OR = 12.98, 95\% CI [8.29, 20.30]$). By contrast, acquaintances virtually never provide instrumental support ($OR < 0.01$). Daily contact frequency is negatively associated with instrumental support ($OR = 0.77, p = 0.005$), reflecting the spatial dispersion of collegiate students from the parental households that provide financial and material support. Long-standing duration (> 10 years: $OR = 1.81, p = 0.035$)

¹Due to extreme dyadic cell sparsity ($n = 1$ acquaintance tie assigned to Instrumental Support), the parameter estimate for acquaintances predicting instrumental support exhibits quasi-complete separation ($OR = 2.91 \times 10^{-6}, SE = 271.99$). To prevent severe logarithmic axis distortion and ensure visual legibility across all well-identified parameters, this degenerate estimate is omitted from Figure 2.

Predictors of Latent Social Support Configurations

Multilevel Logistic GLMM Odds Ratios Sorted by Direction and Effect Size ($N = 22,737$ ties)

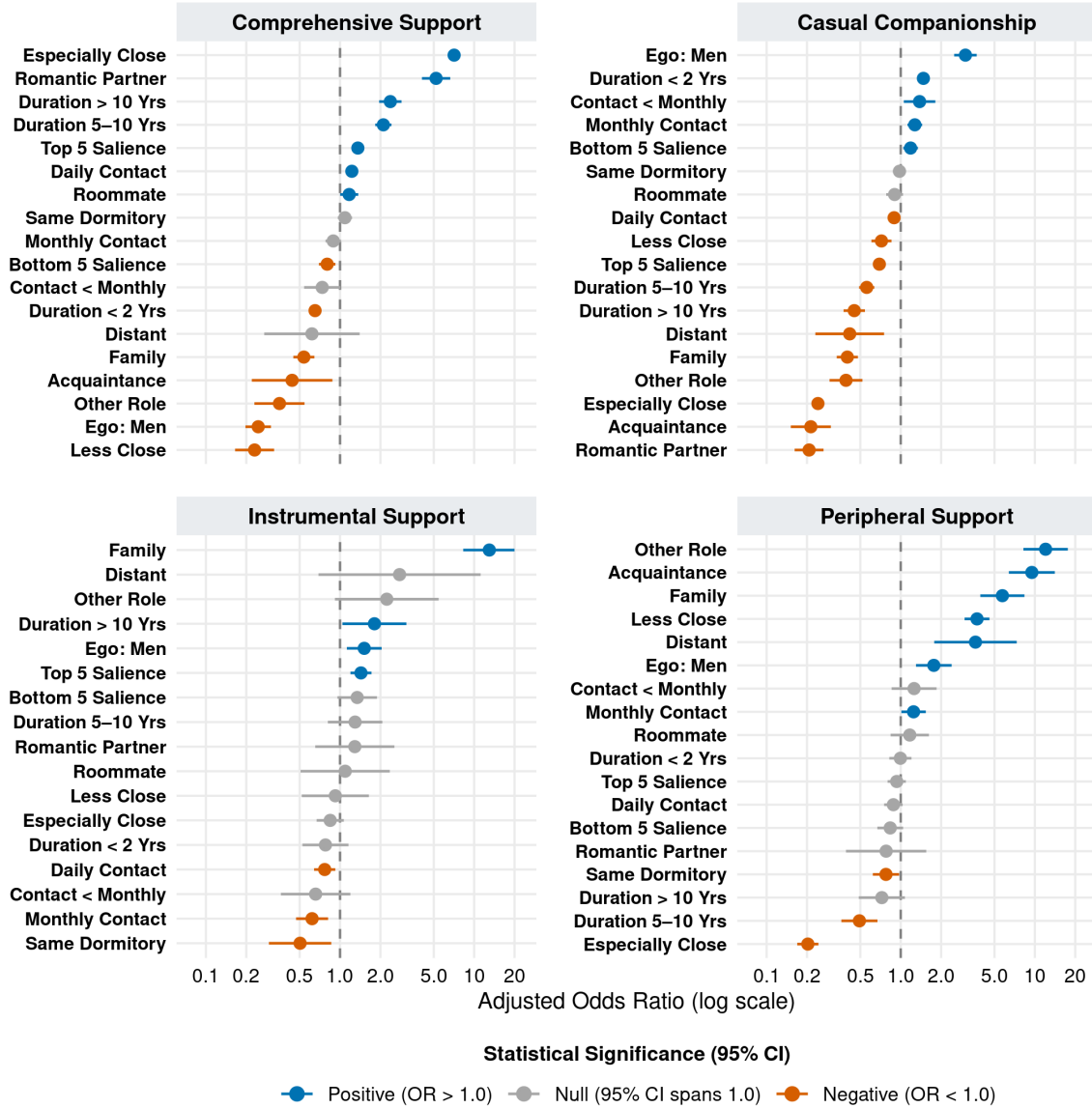


Figure 2: Adjusted Odds Ratios from Multilevel Logistic GLMMs Predicting Latent Social Support Configurations

Note: Forest plot of adjusted odds ratios and 95% confidence intervals estimated from multilevel logistic regression models with random ego intercepts ($N = 22,737$ ties across $N = 581$ egos). The dashed vertical line indicates the null effect ($OR = 1.0$). The horizontal axis is plotted on a logarithmic scale. Within each support configuration panel, predictors are sorted by the direction and magnitude of their adjusted odds ratio. The estimate for acquaintances predicting instrumental support exhibits quasi-complete separation due to cell sparsity ($n = 1$, $SE = 271.99$) and is omitted from the plot to preserve visual scale.

and Top 5 cognitive salience ($OR = 1.43, p < 0.001$) further underscore that instrumental ties represent core familial anchors.

Fourth, Peripheral Support captures relational ties characterized by weak sentiments, formal roles, and minimal resource transmission. Ties evaluated as “Less Close” ($OR = 3.71, p < 0.001$) and “Distant” ($OR = 3.61, p < 0.001$) show nearly fourfold increases in the odds of being peripheral relative to “Close” ties, while “Especially Close” ties are rarely peripheral ($OR = 0.20, p < 0.001$). Role relations exhibit sharp sorting: acquaintances ($OR = 9.50, p < 0.001$) and other non-intimate roles ($OR = 12.03, p < 0.001$) are heavily concentrated in the peripheral tier. Long-term relational duration acts as a protective buffer against marginalization (5–10 years: $OR = 0.49, p < 0.001$).

Finally, respondent gender identity and organizational context reveal pronounced structural differences. Men report substantially lower odds of maintaining ties in Comprehensive Support ($OR = 0.25, p < 0.001$), but more than a threefold increase in the odds of Casual Companionship ties ($OR = 3.04, p < 0.001$), illustrating a stark gendered division where men’s personal communities are disproportionately weighted toward activity-based socializing rather than multiplex expressive confiding. Roommates show modestly higher odds of Comprehensive Support ($OR = 1.17, p = 0.046$), whereas shared dormitory residence reduces the likelihood of ties falling into Instrumental Support ($OR = 0.50, p = 0.012$) or Peripheral Support ($OR = 0.78, p = 0.026$).

6 Discussion and Conclusion

6.1 Summary of Key Results

Taken together, the empirical findings mapped across our latent class and multilevel models provide a comprehensive cartography of how social support exchanges are structured across personal networks. Rather than supporting a singular or reductionist account of tie strength, the results show that personal communities are organized through discrete, functionally specialized relational configurations shaped by the intersection of sentiment frames, behavioral interaction, cognitive salience, relational history, and institutional role relations.

At the broadest level, the evidence firmly supports the multidimensional measurement framework advanced by [Marsden and Campbell \(1984\)](#) and the frame-analytic perspective developed by [Lizardo \(2024\)](#). Far from functioning as an undifferentiated continuum from “weak” to “strong,” tie strength comprises distinct relational attributes that do not collapse into a single linear hierarchy or map neatly onto formal role labels. By decoupling emotional closeness, contact frequency, cognitive salience, and duration from social role categories, our

models show that each dimension of tie strength plays an independent, non-redundant role in sorting ties into supportive regimes. Emotional closeness is an indispensable element in generating comprehensive support, whereas interaction frequency primarily shapes companionship and everyday sociability. Similarly, cognitive salience in the name-generation process reflects psychological prominence that enhances access to substantive assistance, as evidenced by higher reported closeness and frequency.

Furthermore, our findings shed light on the specialized functional division of labor embedded within personal networks. The latent class analysis reveals that supportive ties sort into four discrete configurations: Comprehensive Support (41.5%), Casual Companionship (45.8%), Instrumental Support (4.3%), and Peripheral Support (8.4%). These inductively identified regimes challenge conventional assumptions in social network analysis that support exchanges accumulate along a single cumulative scale. Comprehensive support bundles expressive intimacy, advice, and companionship with occasional material assistance, whereas casual companionship provides sociability while strictly eschewing confiding or financial exchange. Conversely, instrumental ties provide vital material and informational support despite low everyday companionship, and peripheral ties provide minimal assistance. Role relations establish normative boundaries around specific resources: financial assistance and instrumental support remain overwhelmingly anchored in family relationships, whereas companionship is almost exclusively negotiated through peer friendships.

Crucially, the multilevel generalized linear mixed models show how tie strength indicators and role relations independently sort personal ties into these distinct support regimes. First, emotional closeness, relationship longevity, and cognitive salience function as the primary engines driving Comprehensive Support. Ties evaluated as “Especially Close” exhibit a sevenfold increase in the odds of providing comprehensive support, reinforced by long-term duration and placement in the Top 5 retrieval tier. Second, Casual Companionship is overwhelmingly structured by peer friendships and moderate affective intimacy. Ties evaluated as especially close show sharply lower odds of remaining in this class because deep emotional intimacy shifts ties into comprehensive support. Alters nominated in the Top 5 salience tier are likewise less likely to be casual companions, confirming that cognitive retrieval prioritizes more supportive confidants over recreational peers. Third, Instrumental Support is anchored in kinship roles rather than by campus proximity or daily interaction. Family ties exhibit a dramatic thirteen-fold elevation in the odds of providing instrumental support relative to friends, and this form of backing is negatively associated with daily interaction, capturing the spatial dispersion of college students from parental households. Finally, Peripheral Support is concentrated among distant alters, acquaintances, and non-intimate roles, with relationship longevity serving as a protective buffer against marginalization.

Furthermore, respondent gender identity and organizational context reveal pronounced structural sorting. Men report substantially lower odds of maintaining ties in comprehensive support but more than a threefold increase in the odds of casual companionship ties, reflecting a gendered division in which men’s personal communities are disproportionately weighted toward activity-based sociability rather than multiplex expressive confiding. Concurrently, shared residential environments shape tie activation: roommates exhibit higher odds of comprehensive support, while shared dormitory residence decreases the likelihood of ties falling into instrumental or peripheral regimes. In the sections that follow, we address methodological limitations and suggest directions for future inquiry, before turning to the broader theoretical implications for social capital and relational sociology.

6.2 Limitations and Suggestions for Future Work

A central methodological challenge in the empirical study of social networks and social support is the operationalization of supportive exchanges within personal communities ([Wellman and Wortley, 1990](#)). In this study, social support was measured using a dedicated ego-network survey battery that recorded binary incidence across four specific exchange domains: companionship, advice, comfort, and financial assistance. While binary indicators minimize cognitive burden across extensive alter rosters (up to 25 alters per wave), they do not capture the affective intensity, frequency, transaction volume, or qualitative depth of supportive interactions ([Small, 2017](#)). A tie that provides emotional comfort during a single acute crisis is coded identically to one that provides continuous, daily emotional grounding, potentially obscuring meaningful gradations in supportive commitment. Future studies can overcome this constraint by incorporating multi-item scales of supportive intensity, tracking continuous financial and resource flows, and deploying experience sampling methods (ESM) or mobile ecological momentary assessments (EMA) to record the real-time activation, context, and duration of supportive interactions in daily life.

Because this study relies on observational survey data, parameter estimates must be interpreted as conditional associations rather than isolated causal effects. Several potential threats to causal inference warrant careful consideration. First, the relationship among tie-strength indicators, role relations, and support exchanges is susceptible to unmeasured confounding and omitted-variable bias. While our multilevel models adjust for respondent random intercepts, ego gender, and residential proximity, they do not account for unmeasured attributes, personality traits, baseline psychological distress, or dispositional attachment styles. In network theory, personal sociability and psychological orientations operate upstream of tie nomination; thus, part of the sorting into comprehensive support may reflect

unmeasured psychological affinities between ego and alter. Future research can address this limitation by collecting multi-informant dyadic data that measure alter characteristics directly, employing sibling or roommate fixed-effects designs, and incorporating psychological attachment inventories to isolate the independent contribution of relational frames from dispositional tendencies.

Second, observational network panels cannot rule out reciprocal causality and dynamic selection processes. While relationship duration and formal role categories precede current support exchanges, the relationship among emotional closeness, interaction frequency, and support provision is inherently bidirectional. While affective intimacy fosters trust that facilitates confiding and mutual aid, the successful exchange of emotional comfort or financial assistance conversely reinforces affective closeness and increases future interaction. Consequently, the models map the structural equilibrium of personal networks rather than isolated causal treatments. Future investigations can adjudicate these directional dynamics using multi-wave cross-lagged panel models and continuous-time event-history analysis to trace reciprocal feedback loops among support activation, sentiment deepening, and tie persistence over time.

Finally, the empirical scope of this investigation is grounded in a specific institutional setting: an undergraduate cohort at a private Midwestern university followed across early adulthood. This life-course transition is characterized by unique structural conditions, including high residential proximity in campus dormitories, dense peer environments, and physical dispersion from parental households. These structural features naturally accentuate the separation between peer companionship and familial financial backing. While the *NetHealth* panel offers rich multi-wave depth, the relational configurations identified here cannot be assumed to generalize straightforwardly to older adult populations, non-collegiate cohorts, socioeconomically disadvantaged communities, or non-Western kinship systems. In societies where extended kinship networks remain the primary source of everyday sociability, or where economic precarity necessitates shared material survival among non-kin, the social distribution of support configurations may follow different organizational logics. Future comparative research should extend this two-stage analytical strategy to nationally representative ego-network panels, examine personal networks across subsequent life-course stages, and evaluate cross-cultural variations in the boundary between kinship obligation and friendship.

6.3 Theoretical Implications: Decoupling Relational Frames in Social Capital

The empirical findings presented here have direct implications for foundational debates in social capital theory and the sociology of personal relationships. For decades, scholarship on social networks has been organized around contrasting characterizations of tie strength. In classical formulation ([Granovetter, 1973](#)), tie strength is conceptualized as a combination of amount of time, emotional intensity, intimacy, and reciprocal services, commonly operationalized as a single linear continuum from weak to strong. Subsequent theories of social capital ([Coleman, 1988](#); [Lin, 2001](#)) frequently conflated tie strength with social role categories and normative expectations, assuming that strong ties uniformly provide multiplex support while weak ties function primarily as informational bridges.

Our results challenge the assumption that tie strength operates as a unidimensional continuum, showing instead that personal networks are governed by a specialized functional division of labor. Building on [Lizardo \(2024\)](#) and [Marsden and Campbell \(1984\)](#), our findings show that “strong” ties are not all created the same. Emotional closeness serves as the primary gateway for expressive confiding and empathetic comfort, but it does not dictate financial assistance, which remains strictly bound to kinship obligations regardless of contact frequency. Conversely, casual companionship ties thrive on high interaction frequency and shared activities without requiring deep emotional intimacy or confiding. Peripheral and acquaintance ties, far from being defective or deficient relationships, serve vital specialized functions in everyday life ([Small, 2017](#)), providing low-stakes sociability and routine interaction without the heavy normative obligations or emotional vulnerability entailed by comprehensive ties.

These findings speak directly to structural theories of social capital. Access to diverse forms of support does not depend simply on maximizing tie volume or accumulating an undifferentiated stock of strong relationships. Rather, social capital requires maintaining a balanced, diversified portfolio of ties across multiple distinct attributes. Personal well-being and resilience depend on access to specialized relational channels: family ties providing durable instrumental backing during crises, peer friends offering everyday companionship, and intimate confidants providing empathetic comfort and advice. Furthermore, our findings highlight the critical role of cognitive salience and relational duration as independent structural anchors. Long-term duration and cognitive prominence preserve access to supportive resources even when physical proximity and interaction frequency are attenuated by geographic distance or life-course transitions. By decoupling relational frames, future network research can better understand how personal communities sustain well-being across the life course.

References

- Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1):1–48, 2015. doi: 10.18637/jss.v067.i01.
- Stephen P. Borgatti, Ajay Mehra, Daniel J. Brass, and Giuseppe Labianca. Network analysis in the social sciences. *Science*, 323(5916):892–895, 2009. doi: 10.1126/science.1165821.
- James S. Coleman. Social capital in the creation of human capital. *American Journal of Sociology*, 94:S95–S120, 1988. doi: 10.1086/228943.
- Benjamin Cornwell, Edward O. Laumann, and L. Philip Schumm. The social connectedness of older adults: A national profile. *American Sociological Review*, 73(2):185–203, 2008. doi: 10.1177/000312240807300201.
- Martin Elff. *mclogit: Multinomial Logit Models for Categorical Responses and Discrete Choices*, 2025. URL <https://CRAN.R-project.org/package=mclogit>. R package version 0.9.15.
- Scott L. Feld. Social structural determinants of similarity among associates. *American Sociological Review*, 47(6):797–801, 1982. doi: 10.2307/2095213.
- Claude S. Fischer. *To Dwell Among Friends: Personal Networks in Town and City*. University of Chicago Press, Chicago, IL, 1982.
- Mark S. Granovetter. The strength of weak ties. *American Journal of Sociology*, 78(6):1360–1380, 1973. doi: 10.1086/225469.
- James A. Kitts. Beyond networks in structural theories of exchange: Promises from computational social science. In Shane R. Thye and Edward J. Lawler, editors, *Advances in Group Processes*, volume 31, pages 263–298. Emerald Group Publishing Limited, 2014. doi: 10.1108/S0882-614520140000031008.
- Nan Lin. *Social Capital: A Theory of Social Structure and Action*. Cambridge University Press, Cambridge, UK, 2001. doi: 10.1017/CBO9780511815447.
- Drew A. Linzer and Jeffrey B. Lewis. poLCA: An R package for polytomous variable latent class analysis. *Journal of Statistical Software*, 42(10):1–29, 2011. doi: 10.18637/jss.v042.i10.
- Siyuan Liu, David Hachen, Omar Lizardo, Christian Poellabauer, Aaron Striegel, and Tijana Milenković. Network analysis of the NetHealth data: Exploring co-evolution of individuals’

- social network positions and physical activities. *Applied Network Science*, 3(1):45, 2018. doi: 10.1007/s41109-018-0103-2.
- Omar Lizardo. Theorizing the concept of social tie using frames. *Social Networks*, 78:80–91, 2024. doi: 10.1016/j.socnet.2024.01.001.
- Peter V. Marsden and Karen E. Campbell. Measuring tie strength. *Social Forces*, 63(2): 482–501, 1984. doi: 10.2307/2579058.
- R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2026. URL <https://www.R-project.org/>.
- Brandon Sepulvado, Michael Wood, Cheng Wang, Ethan Fridmanski, Matthew Chandler, Omar Lizardo, and David Hachen. Predicting homophily and social network connectivity from dyadic behavioral similarity trajectory clusters. *Social Science Computer Review*, 40(1):186–205, 2020. doi: 10.1177/0894439320923123.
- Mario Luis Small. *Someone To Talk To*. Oxford University Press, New York, NY, 2017. doi: 10.1093/oso/9780190661427.001.0001.
- Jeffrey A. Smith. The continued relevance of ego network data. In Ryan Light and James Moody, editors, *The Oxford Handbook of Social Networks*, pages 170–187. Oxford University Press, Oxford, UK, 2021. doi: 10.1093/oxfordhb/9780190251765.013.15.
- Cheng Wang, Omar Lizardo, and David S. Hachen. Neither influence nor selection: Examining co-evolution of political orientation and social networks in the NetSense and NetHealth studies. *PLOS ONE*, 15(5):e0233458, 2020. doi: 10.1371/journal.pone.0233458.
- Barry Wellman and Scot Wortley. Different strokes from different folks: Community ties and social support. *American Journal of Sociology*, 96(3):558–588, 1990. doi: 10.1086/229572.
- Hadley Wickham. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag, New York, NY, 2016. ISBN 978-3-319-24277-4. doi: 10.1007/978-3-319-24277-4.